

## Homework Assignment 2

1. An ideal gas mixture at  $T = 2000\text{ K}$  has the following composition:

| Species       | H <sub>2</sub> | O <sub>2</sub> | H <sub>2</sub> O | OH    | O     | H     | CO    | CO <sub>2</sub> |
|---------------|----------------|----------------|------------------|-------|-------|-------|-------|-----------------|
| Mole fraction | 0.086          | 0.019          | 0.326            | 0.060 | 0.012 | 0.030 | 0.319 | 0.148           |

The molar specific enthalpies and the constant pressure molar specific heats for the various species at  $T = 2000\text{ K}$  are listed in the table, below.

| Species                 | H <sub>2</sub> | O <sub>2</sub> | H <sub>2</sub> O | OH     | O      | H      | CO     | CO <sub>2</sub> |
|-------------------------|----------------|----------------|------------------|--------|--------|--------|--------|-----------------|
| $\bar{h}$ (kJ/kmol)     | 52968          | 59169          | -169040          | 92761  | 284909 | 253352 | -53804 | -302126         |
| $\bar{c}_p$ (kJ/kmol-K) | 34.236         | 37.788         | 51.143           | 34.635 | 20.819 | 20.786 | 36.721 | 60.433          |

- (a) Calculate, for the mixture, the molar mass ( $\bar{M}_m$ ), the molar specific enthalpy ( $\bar{h}_m$ ), the specific enthalpy ( $h_m$ ), the specific gas constant ( $R_m$ ) and the specific heat ratio ( $\gamma_m$ ). {Partial Ans:  $h_m = -4059\text{ kJ/kg}$ ,  $R_m = 356.3\text{ J/kg-K}$ }
- (b) The mixture is formed from the reaction of a hydrocarbon ( $\text{CH}_{1.96}$ ) and oxygen ( $\text{O}_2$ ). Calculate the number of moles of the hydrocarbon and the oxygen in the reactant mixture. From the result, calculate their mole fractions. {Partial Ans.:  $X_{\text{CH}_{1.96}} = 0.4705$ ,  $X_{\text{O}_2} = 0.5295$ }